

1. Open your desktop or laptop, locate the RAM slots, and remove and re-insert the RAM module to simulate reseating — then document each step you took and any issues you observed during the process.

Step	Action Taken	Observation
1	Shut down the laptop and disconnected the power adapter.	Laptop powered off safely.
2	Opened the back cover using a screwdriver.	RAM slot was visible.
3	Released the side clips and carefully removed the RAM module.	RAM module came out without any damage.
4	Checked the RAM and slot for dust or physical damage.	No dust or damage was observed.
5	Reinserted the RAM module firmly into the slot until the clips locked.	RAM was seated properly.
6	Closed the back cover and powered on the laptop.	Laptop booted normally and detected the RAM correctly.

2.Create a troubleshooting checklist for diagnosing a faulty GPU in a gaming PC, including at least 5 steps from initial symptom identification to confirming the fix.

Hint: Think about display issues, driver checks, and physical connections.

Step	What to Check
1. Identify the Symptoms	Check for a blank screen, no display, screen flickering, graphical artifacts, or game crashes.
2. Check Physical Connections	Ensure the GPU is properly seated in the PCIe slot and all power cables are securely connected.
3. Inspect the Monitor and Cable	Verify that the monitor cable (HDMI/DisplayPort) is connected correctly and test with another cable or monitor if needed.
4. Check Graphics Drivers	Update, reinstall, or roll back the GPU drivers to rule out driver-related issues.
5. Test the GPU	Install the GPU in another PC or test with a different GPU in the same PC to determine whether the graphics card is faulty.
6. Confirm the Fix	After repairing or replacing the GPU, restart the PC and verify that the display works normally without crashes or artifacts.

3. Simulate a storage issue by unplugging and replugging a SATA or NVMe drive (SSD/HDD) on a test system, then describe the error messages or BIOS indicators you see before and after reconnecting.

Step	Observation
1. Shut down the PC and disconnected the power supply.	System powered off safely.
2. Disconnected the SATA/NVMe SSD from the motherboard.	Storage device was removed successfully.
3. Powered on the PC without the storage drive connected.	BIOS did not detect the SSD/HDD and displayed "No Boot Device Found" or "No Bootable Device" .
4. Powered off the PC and reconnected the SATA/NVMe SSD securely.	SSD/HDD was connected properly.
5. Powered on the PC again.	BIOS detected the storage drive and the operating system booted normally.

Error Message / BIOS Indicators

Before Reconnecting:

- No Boot Device Found
- Boot Device Not Detected
- SSD/HDD was **not listed** in the BIOS Storage section.

After Reconnecting:

- SSD/HDD was **detected** in the BIOS.
- The operating system loaded successfully.
- No boot error messages appeared.

4. Given a scenario where a PC randomly restarts, list which component (RAM, GPU, storage) you would check first and explain your reasoning, then outline the specific diagnostic steps you would perform for that component.

PC Randomly Restarts – Diagnostic Plan

Component to Check First: RAM (Memory)

Reason:

RAM is one of the most common causes of random PC restarts. A faulty or loose RAM module can cause system crashes, blue screen errors, and unexpected restarts.

Diagnostic Steps for RAM

Step	Action
1.	Turn off the PC and disconnect the power cable.
2.	Open the PC case and remove the RAM module carefully.
3.	Clean the RAM contacts gently and reinsert the RAM firmly into the memory slot.
4.	If there are multiple RAM modules, test one RAM stick at a time in different memory slots.
5.	Restart the PC and check whether the random restart problem has been resolved.